

MPC-Based Co-Design of a Hybrid Tensegrity Humanoid

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Abstract—This paper examines how articulated joints and tensegrity structure should be combined in a full-size humanoid robot. Three machines of identical anthropometry are constructed in simulation and evaluated under a receding-horizon iLQG controller: a pure tensegrity machine, a fully hinged 27-DoF tendon-driven machine, and a hybrid of the two. A wrench-closure analysis shows that a hinge-free cable joint requires both overlapping cages and counter-wound cable families; without the latter the cable set has no self-stress state and no balance controller can exist. The pure tensegrity machine stands but presents 210 degrees of freedom that no available planner stabilizes. The fully hinged machine walks, but raises peak ground reaction at every drop height relative to a rigid baseline, and its cable network does not meet its own torque specification at gait poses. The hybrid places bearings at hips and knees and tensegrity joints at ankles and waist; at 1.68 m and 20.1 kg it walks at 0.18 m/s in 10/10 of trials with 0.8 kg of articulation hardware. A joint-count study over rigidized variants of the hybrid quantifies the trade-offs: rigidizing the tensegrity joints increases gait speed and payload but removes impact attenuation (−27 to −32% at the ankle joint, at series-elastic cable stiffness) and requires controller retuning, while removing the remaining bearings yields no controllable machine. The results are consistent with an allocation principle in which bearings serve joints with large, pose-varying torque demands and cable interfaces serve joints where impact loads enter the structure.

I. INTRODUCTION

Mass is the binding constraint on humanoid robots. It sets the actuator torques, which set the motor mass, which sets the battery mass, which sets the mass again; it sets the energy of every unintended collision; and it governs the hazard a machine of human size poses to the people around it. Conventional humanoids are chains of rigid links with actuators at the joints, an architecture that concentrates mass distally, makes collisions stiff, and couples structural strength to structural weight.

Tensegrity structures [1], [2] invert this logic: disconnected struts float in a network of pre-tensioned cables, every member is loaded along its axis, and stiffness is a function of prestress rather than of section. These properties have made tensegrity attractive for planetary landers and mobile robots [3], [4], [5], but they have not been carried to a walking machine of human size, and whether they survive that transition is not something statics can answer.

This paper reports one full iteration of a design loop whose inner step is model-predictive control: generate a parametric tensegrity kinematics, test its controllability under tension-only

actuation, size actuators and members from simulated load traces, and ask a receding-horizon iLQG controller [6], [7], [8] to balance and walk it. Every design change is justified by whether it improves the gait. Three machines of identical anthropometry travel through the paper: *pure tensegrity*, *fully hinged*, and the *hybrid* between them that the measurements select.

Findings.

- 1) At 1.68 m the hybrid masses 20.1 kg, against 30–80 kg for size-matched humanoids (Table I); articulation hardware totals 0.8 kg because only eight DoF carry bearings.
- 2) Because the prestressed network is also the transmission, actuators can be mounted proximally at no additional mechanical cost; doing so measurably improves walk success and speed (Sec. V).
- 3) Impact attenuation depends on the joint implementation, not on the architecture alone: as built, the tension network *raises* peak ground reaction, while a hinge-free rebuild lowers it by 27–32% provided cable stiffness remains series-elastic (Sec. VI).
- 4) Prestress does not modulate stiffness at rope modulus, where the geometric stiffness term is 0.2% of the elastic one; antagonist co-contraction changes endpoint stiffness by $8.4\times$.
- 5) Pose-dependent torque authority is the limiting constraint on cable joints: 13 of 27 DoF fall below their torque requirement mid-stride, which motivates retaining bearings at hips and knees (Secs. IV, VIII).

Contributions. (i) A 1.68 m, 20.1 kg *hybrid* humanoid, specified to tube, rope and drive class, walking at 0.18 m/s in 10/10 of trials under receding-horizon iLQG (Sec. VII); (ii) a hinge-free joint geometry — overlapped cages, counter-wound cable families — with the result that force closure alone leaves a joint uncontrollable: full wrench closure ($t_6 > 0$) is required for any balance controller to exist (Sec. VI); (iii) a wrench-closure audit and pose-dependent authority bound applied as a design gate over the gait (Sec. IV); (iv) a joint-count study over rigidized variants of one machine, yielding measured trade-offs and an allocation rule (Sec. VIII); (v) an MPC architecture tracking an analytic gait reference computed inside the cost evaluation, with a pipeline that regenerates every number in this paper from logged runs.

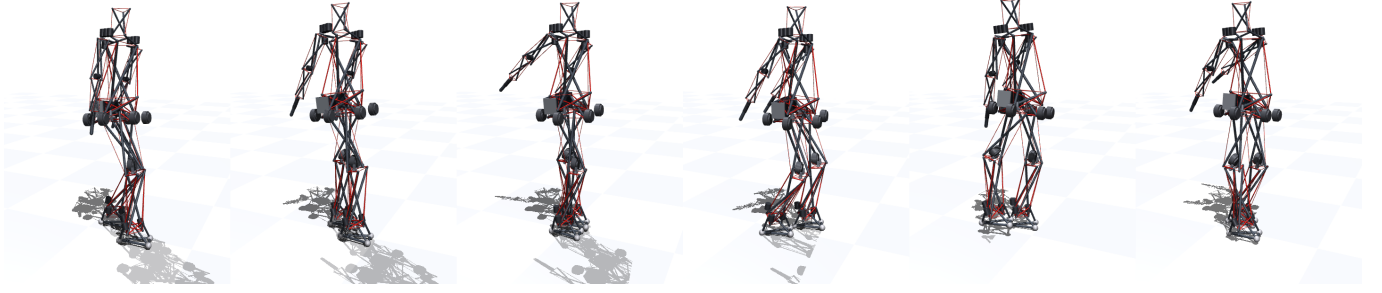


Fig. 1. The hybrid humanoid walking under iLQG in the stride-lengthened configuration of Sec. VII (0.23 m/s, 10/10 of trials); frames advance 7/6 of a gait cycle (phase steps a sixth-cycle per frame). Struts are dark, tension members red; the feet and torso are free bodies held only by counter-wound cable interfaces. Arms hold a servo-referenced posture. 1.68 m, 20.1 kg with both battery packs; see the video attachment.

II. RELATED WORK AND SIZE-MATCHED COMPARISON

Tensegrity robotics. From its artistic origins [1], [2] and structural analysis [9], robotic tensegrity has been demonstrated mainly in rolling or crawling machines: NASA’s SUPERball [3], [4]; the field is surveyed in [5]. Control approaches, including learned ones, are covered by the survey; MPC with inverse-statics tension distribution on tensegrity spines [10] is the closest precedent for our architecture, though the spines are few-DoF modules loaded quasi-statically. Manipulators [11], modular cells [12] and a variable-stiffness spine [13] show limb-like assemblies are feasible; form-finding is reviewed in [14]; biotensegrity reads vertebrate joints the same way [15].

Tendon-driven and compliant humanoids. The JSK musculoskeletal line — Kenshiro [16], Kengoro [17] — routes on the order of a hundred muscles over a human-mimetic skeleton with proximal motors and the same unilateral tension constraint we face. What distinguishes the present design is that the tendons *are* the structure: the network applying joint moments also carries the structural tension, so no separate skeleton exists for the muscles to act on. Joint-level compliance in humanoids has its own lineage through series-elastic actuation [18], [19]; here series elasticity reappears as a requirement on the tension members themselves (Sec. III). Soft exosuits [20] established that textile load paths carry forces of this magnitude, motivating our longer-term goal of mounting the structure in standard apparel.

Co-design. Morphology and control have been optimized jointly by sensitivity analysis through optimal control [21] and by reinforcement learning over design distributions [22]. We report one manual iteration of such a loop, with a predictive controller as the inner step; nothing here searches design space.

MPC for legged robots. iLQG synthesizes whole-body humanoid behavior from cost functions [7], [23], and MuJoCo MPC [8] provides real-time planners. The gait reference draws on Raibert [24] and the capture point [25]. Actuator selection follows Mini Cheetah’s quasi-direct-drive approach [26]. Torque budgets are from Winter [27].

In Table I the mass spread at fixed stature is nearly threefold — architecture, not scale, dominates — and the lightest machines are all tendon-driven with soft shells. The capability

TABLE I
HUMANOIDS OF COMPARABLE STATURE; MASS PER UNIT HEIGHT COMPARES ACROSS SLIGHTLY DIFFERENT STATURES. OUR TWO ROWS ARE SIMULATED MASS BUDGETS, NOT BUILT MACHINES. THE HYBRID’S 46 DOF INCLUDE THE TWO FREE BODIES OF ITS TENSEGRITY JOINTS, DRIVEN BY 30 ACTUATORS.

robot	m	kg	kg/m	DoF	actuation
this work: hybrid (sim)	1.68	20.1	12.0	46	hinge+tensegrity
this work: hinged (sim)	1.65	27.6	16.7	27	tendon, hinged
1X NEO Gamma [28]	1.65	30	18.2	—	tendon, soft shell
HRP-4 [29]	1.51	39	25.8	34	rigid, geared
Unitree G1 [30]	1.32	35	26.5	23	rigid, quasi-direct
Kengoro [17]	1.70	56	32.9	114	musculoskeletal
Fourier GR-1 [31]	1.65	55	33.3	40	rigid, geared
PAL REEM-C [32]	1.65	80	48.5	44	rigid, geared

gap is equally real: Fourier’s GR-1 walks at 1.4 m/s where the hybrid reaches 0.18 m/s in simulation.

III. THE HYBRID DESIGN

The final design (Fig. 2) is 1.68 m sole-to-crown and 20.1 kg, with thigh and shank 0.40 m (Winter anthropometry [27]) and three joint implementations (Fig. 3). *Hips and knees are bearings* (3-DoF gimbal and clevis pins) because they carry the gait’s 40–80 N·m through large excursions. Each knee is an alloy-steel clevis pin in a 61802-class bearing pair (142 g); each hip a three-axis gimbal (257 g); all sized with safety factor 2.3 against the measured 1.5 kN stumble reaction. Articulation hardware totals 0.8 kg against the 1.89 kg of a fully hinged build. *Ankles and waist are tensegrity joints*: each foot is a tri-pad plate (heel and two forefoot corners, the insertion geometry of the biological ankle [15]) whose mast overlaps into the shank cage; the torso is a free body whose cage nests into the pelvis girdle. Both are held only by the counter-wound cable interfaces of Sec. VI: twelve cables each, no bearing, and force and wrench closure by construction. Cables at a hinge-free joint act at the cage radius rather than through a ~ 37 mm effective arm, so the same moment costs an order of magnitude less tension: 0.165 kg XM540-class drives replace the 0.96 kg units a hinged ankle needs. Each arm has seven DoF: a shoulder gimbal, an elbow, and a three-axis wrist, with all drives massed and drawn. While walking, only the two swing DoF per arm are planned; the rest hold position on servo stiffness, modeled explicitly as joint stiffness. The neck

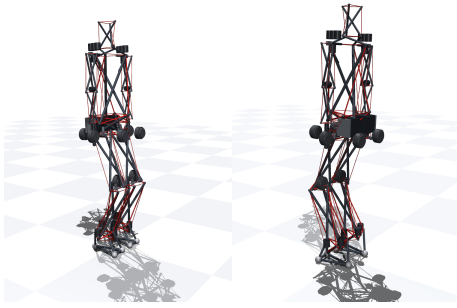


Fig. 2. The hybrid humanoid, front and rear. Every segment is a prism cage; AK70-class hip drives ring the pelvis girdle, battery packs ride posterior rails, XM540-class drives sit on shank rings and torso. Red members are cables: cell cables triangulating each cage, counter-wound joint cables at ankles and waist.



Fig. 3. The three joint types, left to right: hip — three-axis gimbal: yaw pivot descending from the girdle, cross block, roll and pitch axles; knee — drive disc on the thigh cuff, clevis lugs and steel cross-pin between the cages; ankle — no bearing: the foot’s mast reaches into the shank cage and twelve counter-wound cables (red) carry the joint. Cell cables are hidden in the first two panels for clarity.

is rigid in this iteration: the head cage mounts to the torso on a fixed strut tripod, with no articulation. Two 2.5 kg battery packs ride posterior pelvis rails below the waist joint.

Every segment is a Snelson 3-strut prism cell at the *form-found* equilibrium twist. Released free-floating in zero gravity, the cell converges to $29.8^\circ \pm 0.4^\circ$, and that measured twist, not the textbook 30° , generates all cages.

Two earlier variants of the same anthropometry bound the design space and recur as comparison points: the *fully hinged* 27-DoF machine (every joint a bearing, tendon-driven — the configuration most humanoids use) and the *pure tensegrity* machine with no bearings (Sec. VI). Joint torque limits come from human walking moments scaled to robot mass at safety factor ≈ 2 (hip pitch and knee 80 N m, ankle 70 N m, shoulder 30 N m on the hinged machine, 15 N m XM540-class on the hybrid).

Member specification. Since the structural argument for tensegrity rests on the members, each is dimensioned explicitly from the simulated load cases rather than covered by a bulk mass allowance. 60 CFRP tubes, 12–20 mm outer diameter: the governing axial load is the ground reaction entering a cage through its joint (485 N per strut at a 0.1 m stumble), not cable tension. Even so, Euler sizing at safety factor 2 puts *every* strut at the minimum manufacturable wall (0.5 mm; worst margin 6.0 $\times$). The compression network is therefore

TABLE II
MASS BUDGET OF THE HYBRID. EVERY DRIVE IS MOUNTED PROXIMALLY TO THE JOINT IT SERVES; THE TWO TENSEGRITY JOINTS CARRY NO BEARINGS.

item	n	kg	where
hip drives, AK70-class	6	3.1	pelvis girdle
knee drives, AK70-class	2	1.0	thigh cuffs
ankle tension drives, XM540	6	1.0	shank rings
waist tension drives, XM540	6	1.0	torso ring
arm drives, XM540/XM430	14	1.7	torso + arm
articulation hardware (gimbals, clevises)		0.8	hips, knees
battery packs, hot-swappable	2	5.0	pelvis rails
compute + power		0.5	pelvis
structure: cages, cables, feet, masts		6.0	distributed
total		20.1	

manufacturability-limited, not load-limited, at 0.67 kg of tube plus 0.36 kg of fittings. Tendons are 12-strand UHMWPE sized from the larger of the gait’s 99th-percentile tension and the loop requirement $\tau_{\max}/r_{\text{eff}}$ (4 mm legs and waist, 2.5 mm shoulders, 1 mm cell cables; worst RMS utilization 12%). The rope itself is 0.10 kg, but its 444 terminations mass 1.78 kg and the 93 node clusters 0.84 kg — a network of this kind is priced by its interface count, not its cable count.

Cable stiffness: model versus hardware. Specified rope runs 0.3–7.1 MN/m; the simulation models cables at 600–2500 N/m. A deliberate series-elastic element of 20–168 kN/m [18] is required for prestress to be practical to set during assembly, and is what the transmission analysis assumes; the model remains $\sim 46\times$ softer than even that element — the largest modeling gap in this work (Sec. IX).

IV. MODELLING, AUTHORITY, AND CONTROL

The machines are modeled in MuJoCo [33] by a parametric generator that produces, from one description of the node geometry, a *joint-torque* model (motors at the hinges; the planner’s model) and a *cable-driven* model (114 tension-only actuators). Two implementation points matter beyond standard MuJoCo model files. *Cables must be tension-only*: a MuJoCo tendon with a single spring length pushes when short, making every “cable” secretly a strut; all cables therefore use a deadband spring, $F = k \max(0, \ell - \ell_0)$ with ℓ_0 set by the prestress fraction and recomputed from the compiled model so wrapped cables carry the intended prestress.

Long-axis rotations need capstan drums: a cable helixed around a limb has a 16–25 mm moment arm about the limb axis, so yaw DoF carry a coaxial drum (35–55 mm) that fixes the arm independently of cage geometry. Four checks run automatically after every geometry change: all 372 cable endpoints coincide with strut tips (no cables ending in mid-air), the authority analysis below, a scripted balance test, and the same balance by pure tension distribution on the cable model at 250 Hz.

Cable authority. With tension-only actuation the achievable

TABLE III
CABLE AUTHORITY (EQ. 1) ON THE FULLY HINGED VARIANT DEPENDS ON WHERE IT IS EVALUATED AND WHICH CABLES A MOTOR CAN REACH. ENTRIES: DoF SHORT OF THEIR TORQUE LIMIT, AND WORST MARGIN OVER 27 DoF.

configuration	actuation set	DoF short	worst margin
assembly pose	all cables	0/27	$1.01\times$
gait poses	all cables	10/27	$0.00\times$
assembly pose	36 loops	6/27	$0.66\times$
gait poses	36 loops	13/27	$0.00\times$

torque about DoF j is bounded by

$$\tau_j^+ = \sum_c \max(0, M_{jc}) \bar{F}_c, \quad \tau_j^- = \sum_c \min(0, M_{jc}) \bar{F}_c, \quad (1)$$

where $M_{jc} = \partial \ell_c / \partial q_j$ is the moment arm of cable c about joint j and $\bar{F}_c$ its tension limit — the per-DoF projection of the wrench-feasibility condition of the cable-driven parallel robot literature [34], [35]; we claim no novelty for the bound, only for applying it to a legged machine as a requirement *over the whole trajectory* rather than at one pose. Used this way it caught three errors: hip yaw and wrist had exactly zero authority in one direction because all cable families wound with the same handedness (opposite-handed winding is required for bidirectional yaw — a rule that returns, generalized, in Sec. VI); shoulder roll had $0.15\times$ the required torque from a misplaced socket; and ankle cables saturated during push recovery until the foot nodes moved to heel/forefoot positions. As a guarantee it is far weaker than the assembly-pose number suggests (Table III): at assembly, all 27 DoF are bidirectional (worst margin $1.01\times$); at logged gait poses, 10 of 27 fall below their torque limit (hip pitch $0.87\times$, knee $0.61\times$), and the 72-cable, 36-motor bill of materials leaves 13 of 27 short. The exact zeros in Table III are the capstan drums: wrap disengages at some wrist and shoulder angles, and authority there is not merely small but absent. Covering 80 N m at the 1.5 kN cable limit needs $r_{\text{eff}} \geq 53.3\text{ mm}$; the cage offers 37.5 mm at hip pitch. This shortfall is the quantitative reason the hybrid keeps bearings at hips and knees. The moment arms collapse mid-stride, and adding motors does not help: a second cable of the same family loses its arm at the same poses.

Predictive control. The controller is MJPC’s asynchronous iLQG [8], [7] with control limits [23]: horizon 0.6 s , planning timestep 20 ms , plant at 4 ms . Hand-designed cost terms alone produced only stepping in place; two of those terms actively opposed walking (a pelvis line term forbids the weight shift that unloads a foot; a permanent capture-point term is wrong in single support). The remedy is an *analytic gait reference computed inside the cost evaluation*: gait phase is a function of time alone, and the pelvis tracks a lateral sway timed so that weight is over each stance foot just before the other lifts. Stance feet slide back while swing feet return with sinusoidal clearance, half a cycle apart. The swing landing point is biased by center-of-mass velocity error in the manner of Raibert [24] and the capture point [25]; this bias is what makes stepping stabilizing. Foot targets become leg-joint references by ana-

lytic flat-foot inverse kinematics, with posture referenced to a bent-knee walk-ready pose.

Walking results, fully hinged variant. This machine validates the pipeline and produces the load traces used for dimensioning: $0.21 \pm 0.01\text{ m/s}$ in 7/10 of ten trials, longest run 6 m in 29 s , at $0.60\times$ real time on a laptop CPU. Cost of transport is 3.3 . Drives are dimensioned from the traces: force bandwidth through series compliance $f_{\text{bw}} = \frac{1}{2\pi} \sqrt{k_{\text{ser}}/m_{\text{eff}}}$ with $m_{\text{eff}} = I_r G^2 / r^2$ splits the transmissions by what dominates each joint: spools where stroke dominates (hips, knees, waist), lever arms where bandwidth dominates (ankles, 46 Hz), and drums where the rotation axis itself is the problem (yaw). The resulting 36-motor bill of materials, however, *does not meet its own torque specification*: a single loop delivers 56 N m at hip pitch against 80 required, and Table III shows the shortfall widening over the gait. The fix is geometric (enlarge r_{eff}), not more motors.

V. EVIDENCE ON THE FULLY HINGED VARIANT

Five experiments run against a *rigid-equivalent baseline* — the same 27-DoF model with the tension network removed — with actuation held constant, so only the structural paradigm varies (Table IV).

E1, impact. In matched-pair drops of $0.05\text{--}1\text{ m}$ the tension network *raises* peak ground reaction at every height: $+14\%$ limp at 0.1 m , and $+110\text{--}197\%$ at physical cable stiffness. A bearing puts the elastic network in parallel with a stiff path, and impulses take the stiff path. **E2, stiffness.** Under a 200 mN differential endpoint probe, cables raise lateral stiffness $2.4\times$ by their presence alone, but assembly prestress swept $0\text{--}5\times$ moves it only within $11.6\text{--}16.6\text{ N/m}$: prestress does not modulate stiffness at rope modulus (Sec. VI isolates why). Differential tension does — shoulder co-contraction moves $14\text{--}115\text{ N/m}$ ($8.4\times$) with no added hardware. **E3, degradation.** All 15/15 single cable cuts and every random loss up to $k = 8$ survive standing plus a push; an adversary selecting against the authority margin kills at $k = 8$. No single cable is structural, where one failed gearbox in a rigid robot is one dead joint. **E4, proximal mass.** Moving motor masses to the joints they drive, totals identical, degrades walking from 7/10 to 6/10 and 0.21 to 0.17 m/s , quantifying the benefit of proximal mounting under tendon transmission. **E5, tendon realisability.** Mapping logged walking torques through the bounded tension distribution: the network reproduces demanded torques exactly for most of the stride (median residual $< 10^{-3}\text{ N m}$) with concentrated failures — a cable pinned at its 1.5 kN limit in 10.9% of states, leaving up to 39 N m undelivered; the gait asks for 2407 N peak tension, $1.6\times$ the specified rating. Re-planning *inside* the achievable envelope separates topology from actuation: a gait exists for the full network (7/10) but not for the bill of materials (0/10). The topology can support a gait; the actuation set as specified cannot deliver one. Its capability envelope (65% speed tracking to 0.29 m/s ; trunk payload flat to 20 kg) appears as the grey references in Fig. 4.

TABLE IV
EXPERIMENT SUMMARY, ALL ON THE FULLY HINGED VARIANT AGAINST
THE RIGID-EQUIVALENT BASELINE, ACTUATION HELD CONSTANT;
WALKING CONDITIONS $n = 10$.

id	property	result
E1	impact tolerance	network raises peak reaction at every drop height (+110+197% at physical stiffness)
E2	stiffness on demand	assembly prestress: no effect (11.6–16.6 N/m); co-contraction: $8.4\times$
E3	graceful degradation	all singles and random $k \leq 8$ cuts survive; adversarial $k=8$ fails
E4	proximal mass	distal motors: 6/10 vs 7/10 walks, 0.17 vs 0.21 m/s
E5	tendon realisability	13/27 DoF short over gait; gait exists for full network, not for BOM

VI. THE HINGE-FREE JOINT

Is the fully hinged variant’s failure to deliver impact tolerance and prestress-tunable stiffness a fact about tensegrity, or about its hinges? An audit answers the prior question first: that machine’s joints *are* bearings — every articulated interface fixes all three relative translations, and the wrench-closure LP (closure iff $\max\{t : W\lambda = 0, \lambda \geq t\mathbf{1}\} > 0$ for the matrix W of unit-tension cable wrenches [34], [35]) reports 0/14 joints able to resist even pure forces by cable alone: every crossing cable pulls the segments together, and nothing resists approach. Removing the hinges confirms it dynamically — the robot telescopes and collapses within two seconds.

Force closure requires overlap; wrench closure requires counter-winding. In a tensegrity tower, stages *overlap*: the distal ring sits inside the proximal cage, giving the interface a downward as well as an upward cable family. On an isolated two-cage testbed (fixed proximal cage, three floating distal struts, 1 kg payload), force closure appears from $0.2H$ of overlap and the joint carries load. Force closure is not enough. An earlier version of this analysis read the wrench-closure margin $t_6 = 0$ as benign; controlling the joint exposed the error. $t_6 = 0$ means the cable set has *no self-stress*: no co-contraction state and zero moment authority about some axis. Tension-only actuators can then only add load to an equilibrium, never oppose a disturbance, so no balance controller can exist. A controllable joint needs $t_6 > 0$, with actuators choosing how much moment to exert, including none. The single cross family winds one way and cannot do it; adding the *counter-wound mirror family* closes the wrench set at the same cable count. This is the opposite-handedness rule required at the yaw drums, not previously recognized as general. With the counter-wound pair the joint carries load, holds ≥ 1.3 kN of co-contraction authority, and a chain of them stands passively.

Impact attenuation at the hinge-free joint. Dropping the testbed against a mass-matched control whose distal cage rides a ball joint, peak ground reaction falls from 287 to 209 N at 0.05 m, 404 to 282 N at 0.10 m, and 568 to 388 N at 0.20 m — attenuation of 27–32% at every height, where the hinged humanoid *raised* it (E1). Remove the bearing and the elastic network is in series with the load rather than parallel to a stiff path, the arrangement the tensegrity-lander results rely

on [3]. The attenuation belongs to the series compliance that the topology puts into the load path, not to the topology alone: repeating the drops at $10\times$ cable stiffness (400 kN/m, toward bare rope) *inverts* the comparison, the cable joint landing +5 to +21% harder than the pin joint. Hinge-free geometry and series-elastic cables buy impact tolerance jointly; neither suffices alone. **Prestress-tunable stiffness remains absent**, and the mechanism is a materials condition rather than a topology property: tangent stiffness is an elastic term $\mathcal{O}(EA/L)$ plus a geometric term $\mathcal{O}(T/L)$, and prestress tunes stiffness only when they are comparable. At the rope modulus this design specifies the geometric term is 0.2% of the elastic one (hinge or no hinge); the cell-scale demonstrations that motivated the claim used cables three orders of magnitude softer. Variable stiffness must be budgeted as an actuation cost (co-contraction, E2), not an assembly parameter.

Composition into a complete assembly. Generating the entire machine as a graph of overlapping cells — no bearings, nothing touching — stands (66 struts, 306 cables; limb chains droop ≈ 5.9 mm/cell). But junctions cannot be cells (a prism hosts two interfaces; a pelvis needs three), so pelvis and torso become rigid strut clusters, and carrying the 22 kg of motors and batteries takes 1000 N of prestress per cable — an order of magnitude beyond the single-joint level, forcing 3 kN-rated cell cables. The pure machine exists structurally, at a quantified cost; what does not exist is a controller for its 210 floating DoF (Sec. IX).

VII. THE HYBRID WALKER

The prescription that emerges — bearings where torque requirements dominate, tensegrity where compliance dominates — is the hybrid of Sec. III. This section reports its walking performance and the whole-machine effects of its two tensegrity joints.

Gait synthesis. The controller is the analytic-reference iLQG task with the ankle posture rows deleted. With the ankle cables *passive*, every configuration fails identically: the planner does not lift the swing foot. Without ankle actuation there is no center-of-pressure control, so a foot can be unloaded only by whole-body sway, and lifting a loaded foot would lift the robot. Welding the feet to the shanks walks immediately at 0.18 m/s, isolating the cause. Giving the planner tension actuators on two of the four ankle cable families (twelve controls, tension-only) recovers walking at 0.18 m/s in 10/10 of trials — at a running cost *below* the welded control, because the planner exploits the compliance it now commands: one family lifts the hanging foot for swing clearance, the other pitches the toe, differential tension steers the landing. With actuated cables the tensegrity joint therefore constitutes the better-performing ankle in this comparison. Actuator utilization confirms the sizing: hip and knee motors average 9% of their limits, ankle actuators 7% with 250 N peaks at foot lift; the 46 Hz bandwidth demand of the hinged ankle disappears, absorbed by cable compliance. The torso rides the waist cables within $\pm 2^\circ$, and pulling one battery mid-stance leaves the gait intact.

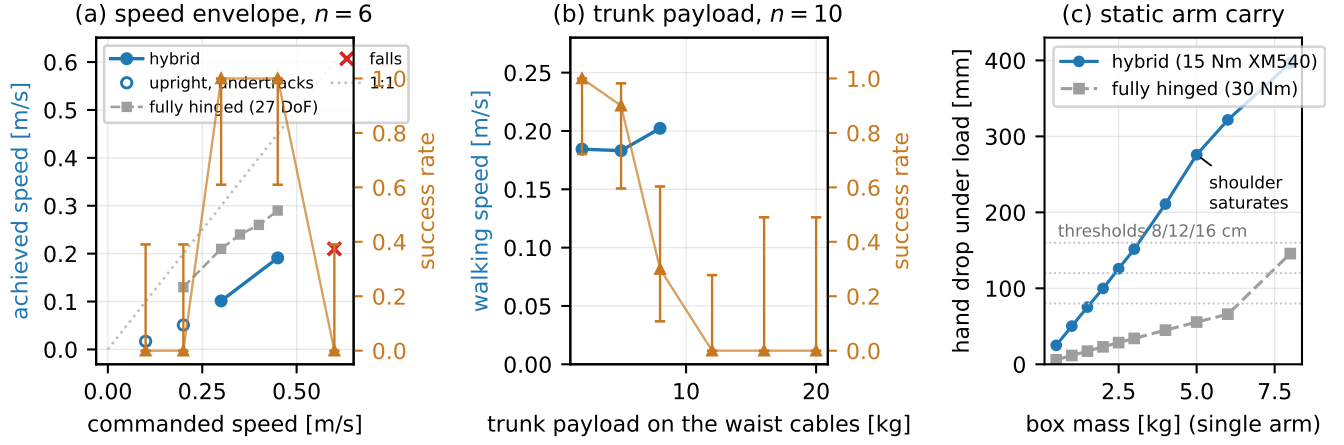


Fig. 4. Capability sweeps on the hybrid; error bars are Wilson 95% intervals on the success rate. (a) Achieved speed and success rate vs. commanded speed ($n = 6$), fully hinged envelope in grey; 0.10–0.20 m/s commands are upright but undertracking, not falls. (b) Trunk payload carried above the tensegrity waist ($n = 10$ to 12 kg, $n = 4$ beyond): reliable at 2 kg, degrading through 8 kg, gone by 12 kg for a centered load, where the hinged machine’s rigid trunk was flat to 20 kg (a placement intervention restores 10 kg at 10/10; see text). (c) Static single-arm carry, matched protocol (pelvis welded, identical pose and gains): hand drop vs. box mass, thresholds dotted. The hybrid is sag-limited (53 mm/kg; hold moves 1.5→3.0 kg with the threshold), the hinged machine motor-limited (16 mm/kg until its 30 N m shoulder saturates; 6.0→8.0 kg).

Capability envelope (Fig. 4). *Speed*: the hybrid has a reliable command band rather than a smooth degradation: 6/6 at a commanded 0.30 m/s (0.10 m/s achieved), 6/6 at 0.45 (0.19), and falls in every trial at 0.60. Below the band it does not fall but undertracks (0.02–0.05 m/s at 0.10–0.20 commanded): the sway-and-step reference has a minimum viable tempo. Gait-parameter retuning then extends the band. Lengthening the stride (cadence 1.1→0.8 Hz at the same command) and raising the swing-height reference to 0.24 m with doubled cartesian tracking weight reaches 0.23 m/s at 10/10, with measured swing clearance of 42 mm — the configuration of Fig. 1. The swing-height reference must be commanded at roughly three times the desired clearance: clearance tracks the reference (11, 18, 42 mm at references of 0.08, 0.16, 0.24 m) while extending the planning horizon has no effect. Raising the speed command to 0.55 at the original cadence reaches 0.25 m/s at only 3/6. The remaining panels of Fig. 4 are measured at the pre-retune configuration. Tracking is the larger difference, $\approx 42\%$ of command against the hinged 65%: part of every push-off is stored in the elastic network rather than delivered to the ground, a speed cost of compliance that Sec. VIII measures joint by joint. *Trunk payload*: boxes on the torso ride *above* the tensegrity waist, so every added kilogram loads the twelve waist cables. The envelope is correspondingly smaller: 10/10 at 2 kg (speed within noise of unloaded), 9/10 at 5, 3/10 at 8, 0/10 at 12 kg — against flat-to-20 kg on the rigid trunk. At 12 kg every trial falls within the first strides, seven of ten *backward* — the loaded waist pitches the trunk aft faster than the planner corrects within its horizon. The hinged machine’s payload was tendon-limited at the leg; the hybrid’s is limited by pitch stability at the compliant waist. An intervention study at 10 kg identifies payload placement as the effective lever: with the load’s center of mass 4 cm anterior of the waist axis (cancelling the static moment of the posterior

battery packs) and the speed command reduced to 0.40 m/s, the machine carries 10 kg at 10/10 and 0.19 m/s. Waist stiffening and torso-height corrections did not help: the constraint is dynamic pitch balance, not static capacity. *Arm carry*: under one protocol on both machines (pelvis welded, same pose and gains; failure = hand drop from the settled unloaded position), the hybrid holds 2.0 kg at a 12 cm threshold where the hinged machine holds 6.0 kg. The failure modes differ in kind: the hybrid’s limit moves 1.5→3.0 kg as the threshold moves 8→16 cm (sag-limited: 53 mm/kg through seven servo-compliant joints, shoulder at 71% at failure), while the hinged machine’s barely moves, 6.0→8.0 kg (motor-limited: its stiff chain sags 16 mm/kg until the 30 N m shoulder saturates). Halving the drive class costs two thirds of the carry rather than half (compliance compounds the torque deficit), and the carry loads a path the hinged machine lacked: the busiest waist cable rises from 52 to 88 N under a 2 kg hold.

VIII. HOW MANY JOINTS DOES A HUMANOID NEED?

The hybrid fixes one point on an axis from no bearings to a bearing at every DoF. Table V walks the axis with variants of the *same* machine — identical masses, drives, anthropometry — differing only in which joints are bearings, run under the hybrid’s final controller configuration, bracketed by the two endpoints with their own tuning.

The case for retaining bearings. *Load path*: without overlap and counter-winding a hinge-free joint is not a structure. *Torque*: a bearing delivers full torque at every angle; a cable joint’s authority collapses mid-stride (Table III). *Planning*, in practice the dominant layer: every bearing removes three to five DoF from the state a predictive controller reasons over. The hybrid’s planner handles 46 DoF at $0.3\times$ real time; the pure machine has 210, and no controller in this work or, to our knowledge, elsewhere stabilizes it. Bearings at hips and knees

TABLE V

THE JOINT-COUNT AXIS. MIDDLE ROWS ARE VARIANTS OF THE HYBRID UNDER ITS FINAL CONTROLLER ($n = 5$); ENDPOINTS CARRY THEIR OWN TUNING. “T-JOINTS” COUNTS TENSEGRITY (BEARING-FREE) JOINTS. THE IMPACT FIGURES ARE FROM THE ANKLE-JOINT TESTBED (SEC. VI) AND E1 RESPECTIVELY.

variant	brgs	T-jts	walks	m/s	note
pure tensegrity	0	all	—	—	stands only; no planner
hybrid (design)	0.8 kg	3	10/10	0.18	impact $-27..32\%$ (ankle)
+ rigid waist	0.8 kg	2	0/5 [†]	—	tuning mismatch
+ rigid ankles	0.8 kg	1	5/5	0.21	
+ both rigid	0.8 kg	0	5/5	0.23	
fully hinged (27 DoF)	1.9 kg	0	7/10 [‡]	0.21	impacts $+14..154\%$ (E1)

[†]fails under the hybrid’s tuning; retuned — one parameter, the torso height target, $1.20 \rightarrow 1.28$ (the welded torso stands taller than the cable-mounted one settles) — the same joint set with the same arms walks 4/5 at 0.18 m/s. [‡]own tuning at its 0.30 m/s operating point; its envelope edge is 0.29 m/s at a 0.45 command (Fig. 4a). Source of the load traces and E1–E5.

make the machine *plannable* and its gait torques available at every pose.

Consequences of replacing bearings with cable interfaces. Replacing a bearing with a cable interface provides impact attenuation ($-27..32\%$ at the ankle testbed; $\pm 2^\circ$ torso compliance), an order of magnitude lighter actuation (0.165 vs. 0.96 kg drives at cage-radius moment arms), graceful degradation (no single cable is structural), and robustness to mass change (the battery-removal test). The corresponding costs are *speed* (rigidizing raises gait speed monotonically, $0.18 \rightarrow 0.21 \rightarrow 0.23$ m/s under one controller, because part of each push-off is stored elastically), *planner load* (each cable joint returns six DoF; $0.3\times$ real time vs. the hinged machine’s $0.6\times$), *payload* (the waist ceiling of Fig. 4b), and *tuning fragility*, most visible in Table V: the gait tuned for the compliant waist fails outright (0/5) when that joint is rigidized. Joint compliance is not a drop-in property; controller and structure must be tuned jointly.

Allocation rule. Place bearings where the gait’s torque requirements concentrate, and cable interfaces where impacts enter the structure. Hips and knees carry the largest, most pose-varied torques and no first impact: bearings. Ankles meet the ground first and need an order of magnitude less moment: cables. The waist is the marginal case: a cable waist provides torso compliance at a cost in speed, payload and tuning effort. The hybrid’s split saves 1.1 kg of bearings over the fully hinged machine, recovers the impact attenuation and degradation tolerance that machine lacks, and remains controllable.

The biological counterexample. The human body appears to contradict this rule. Though jointed at every limb, biotensegrity reads the skeleton as a continuous tension network whose articular surfaces are held by ligament and muscle tension rather than axles [15]: the joint sits inside the tensegrity. The biological knee addresses both problems that led this design to bearings: the patella enlarges the extensor moment arm where the cages fall short of $r_{\text{eff}} \geq 53.3$ mm, and graded co-contraction supplies closure and disturbance rejection. Biology adds the controller for the high-dimensional compliant state

TABLE VI

STAGED VALIDATION LADDER: BENCH-SCALE UNITS IN BUILD ORDER, EACH WITH THE SIMULATED PREDICTION A BENCH TEST WOULD CHECK. THE ARM ROW USES THE PROTOCOL OF FIG. 4C.

stage (unit)	quantity	prediction
cell (3 tubes, 9 cables)	twist; stiffness vs. prestress	$29.8^\circ \pm 0.4^\circ$; flat
ankle (plate, mast, 12 cables, 2 drives)	arms; N m/100 N drop at 40 / 400 kN/m	55–66 mm; 16 $-27..32\%$ / inverts
knee pin, hip gimbal	1.5 kN proof load	SF 2.3, no yield
arm (7 DoF, mount)	loaded sag; first mode	53 mm/kg; 3.6 Hz
waist (torso on pelvis)	sink; capacity	24 mm/100 N; holds 200, fails 300 N
leg (guided-pelvis jig)	cable T vs. load	41 $\rightarrow$ 55 N at +200 N
machine	stand, then walk	10/10 at 0.23 m/s

that our planner does not handle. The rule is thus a statement about present-day drives and planners; the human body indicates the all-tensegrity endpoint is reachable with sufficient control.

IX. DISCUSSION AND OPEN QUESTIONS

Measured benefits of the tensegrity paradigm. The paradigm provides mass efficiency (compression network at minimum wall, 1.03 kg), a transmission at no additional mechanical cost (the prestressed network *is* the transmission; E4), redundancy (E3), impact attenuation in hinge-free form only (Sec. VI), hence tensegrity at ankles and waist, and variable stiffness as a co-contraction cost rather than an assembly parameter. It does not provide pose-independent torque, which binds hardest at hips and knees.

Realizability. A simulation-only study must state how its claims can be tested in hardware. Every member and drive is a catalog part with a stated rating, and the machine decomposes into bench-scale units. Table VI attaches to each unit a prediction simulated under the conditions the bench would apply; discrepancies then localize at components rather than inside a 46-DoF machine. Geometric predictions (twist, closure, moment arms) do not depend on the largest model gap, cable stiffness; the compliance predictions do, and sit early in the ladder where refitting is cheapest. Each stage tests the simulation before the next is built.

Limitations. Everything is simulated: no tendon friction, hysteresis, creep or cable mass, and cables $\sim 46\times$ softer than the specified series-elastic element. The joint-scale impact result is stiffness-swept (it inverts at $10\times$); the whole-machine compliance numbers are not and are the most exposed. Cost of transport (3.3) is $5\text{--}15\times$ that of comparable machines and the planner runs at $0.3\times$ real time. Mass parity against an equivalent rigid frame, contact safety, and fatigue remain untested.

Open questions. *Can a hinge-free humanoid be planned for?* The structure exists (Sec. VI) and biology indicates the control problem is solvable; whether by predictive control over a ~ 210 -DoF state or a reduced cell-as-joint model remains open. *Which design variables dominate the next iteration?* Moment-arm geometry (Eq. (1) as a workspace constraint) and tension-member softness, whose trade shows a cliff between 40 and 400 kN/m.

X. CONCLUSION

This paper quantified the trade-off between articulated joints and tensegrity structure within one design family. A humanoid made only of tensegrity cells stands, given overlapped cages and counter-wound cable families, but presents 210 DoF that no planner stabilizes. A fully hinged variant walks, but raises rather than attenuates impact loads, and its cable network does not meet its own torque specification at gait poses. Between them, a 1.68 m, 20.1 kg hybrid with bearings at the eight DoF carrying the gait's torque and tensegrity joints where impacts enter walks at 0.18 m/s in 10/10 of trials, the lightest per unit height in Table I. Rigidizing its tensegrity joints increases speed and reduces compliance; removing its bearings removes controllability. The allocation principle (bearings where torque requirements concentrate, cable interfaces where impacts enter) and the reported trade data constitute one iteration of an MPC-in-the-loop design process and identify the parameters for the next.

REPRODUCIBILITY

All code (generator, MJPC task, experiments, analysis) will be public on acceptance; the video attachment shows the gait. One script regenerates every figure and quantity from logged runs.

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